

A STARTUP BURNS \$50K/MO ON GPT-5.

80% OF THEIR QUERIES ARE SIMPLE FAQs.

"That's an engineer's salary. Up in smoke."

For what? A question a 1.5B model could answer in 300ms.

THE PROBLEM

THE AI INDUSTRY HAS A ROUTING PROBLEM.

30+

NEW MODELS PER MONTH

No team can evaluate them all. So everyone picks one and never revisits.

50–90%

WASTED INFERENCE SPEND

Enterprise budgets burned on over-provisioned models. Money that could hire engineers.

180×

ENERGY PER PREMIUM QUERY

39 Wh vs 0.22 Wh. AI on track for 1,000 TWh by 2026 — mostly waste.

WHO IT HURTS

Developers

10-30x OVERPAY

Enterprises

COMPLIANCE RISK

End Users

SLOW RESPONSES

The Planet

415 TWh / YR

YOUR AI IS PAYING THE TOLL.
For nothing.

THE FIX

ONE LINE OF CODE.

We handle everything else.

BEFORE

STATUS QUO

```
# Same premium model. Every request. Forever. from openai import
OpenAI client = OpenAI( base_url="https://api.openai.com/v1")
response = client.chat.completions.create( model="gpt-5-4", # ←
HARDCODED messages=messages)
```

AFTER MODELROUTER

DROP-IN

```
# Right model. Every request. Automatically. from openai import
OpenAI client = OpenAI( base_url="https://router.acme/v1") # ← ONE
CHANGE response = client.chat.completions.create( model="auto", # ←
ModelRouter decides messages=messages)
```

API COMPATIBLE

Full OpenAI spec. Zero rewrites. Drop-in replacement.

ALWAYS OPTIMAL

Cheapest capable model per request. Automatically.

CONTRACT-AWARE

Upload SLA docs → routing rules extracted. Per request.

ARCHITECTURE

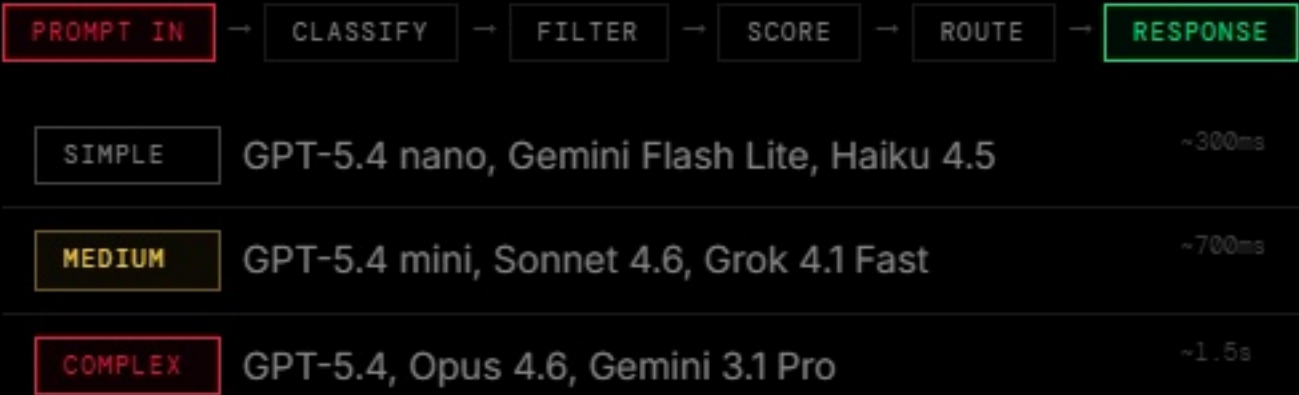
TWO PHASES. FULLY AUTOMATED.

Zero ongoing effort.

PHASE 1 - ONBOARDING = 30s

- **Upload Contract**
PDF or text — SLA, privacy, custom rules
- **LLM Extraction**
Frontier model reads and parses all constraints
- **Customer AI Profile**
Region, providers, latency target, routing tiers
- **Endpoint Live**
/{customer}/v1/chat/completions — ready

PHASE 2 - RUNTIME = 50ms



Classifier: Arch-Router-1.5B · 62ms on GPU · heuristic fallback

FINE-TUNED 1.5B ROUTER — 62MS CLASSIFICATION

TRAINING PIPELINE

Qwen/Qwen2.5-1.5B-Instruct — Base

↓ Katanemo

katanemo/Arch-Router-1.5B — Intent router

↓ GRPO FINE-TUNE — OUR WORK

ModelRouter-Router

↓ GGUF Q8_0

ModelRouter.Q8_0.gguf — 1.6 GB, production

BENCHMARKS

SIMPLE

81.8%

Stock: 87.9%

MEDIUM

85.7%

Stock: 14.3% +71.4pp

COMPLEX

85.7%

Stock: 100%

Overall Accuracy

83.3%

TECH STACK

FastAPI

Next.js

Arch-Router-1.5B

OpenRouter

GRPO

Docker

SQLite

GGUF

Python

TypeScript

Cost Savings vs Premium

59%

Method

GRPO + Unsloth

LoRA Rank

32 (2.3%)

Training

150 steps

Time

2.5 min

Hardware

RTX 3080 8GB

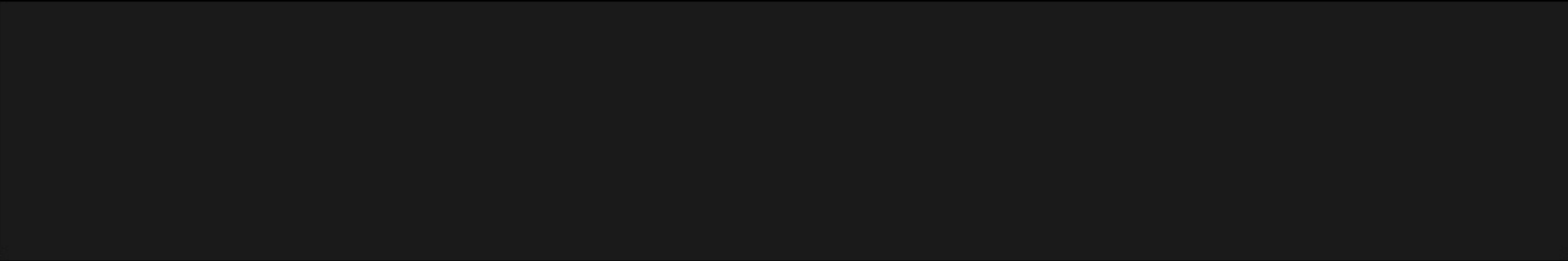
Dataset

172 prompts

IMPACT

ONE CHANGE. BENEFITS AT EVERY LAYER.

LAYER	BENEFIT	HEADLINE
BUSINESS	Eliminates 60–98% of AI inference spend through intelligent right-sizing	60–98% SAVED
USERS	Simple queries → fast answers. Complex queries → powerful models.	RIGHT MODEL
DATA CENTERS	Less unnecessary load on premium GPU clusters. Resources freed for real work.	LESS WASTE
ENVIRONMENT	Up to 180× less energy per routed-away premium call. Thousands of calls saved.	180× LESS
COMPLIANCE	Contract constraints → routing rules. Enforced per request, automatically.	ZERO ERRORS



FIVE STEPS. ONE CONTRACT. IMMEDIATE ROUTING.

01**Onboard ACME**

Region: EU-only from GDPR
DeepSeek blocked
Latency: 1000ms

02**Copy Endpoint**

/acme/v1/chat/completions
Drop into any OpenAI client
Zero other code changes

03**Simple Query**

"What is your return policy?"
→ **SIMPLE**
→ GPT-5.4 nano · 300ms

04**Complex Query**

"Analyze liability exposure..."
→ **COMPLEX**
→ Opus 4.6 · Same endpoint

05 — Dashboard

Model distribution · Cost savings 59% · Live request feed · Per-request explanations

LIVE

THANK YOU.

TEAM DYNAMIC IP

BACKEND & AI

Vardhan

FRONTEND & PLATFORM

Rajesh

"One URL change. Every model. Optimal choice. Zero code changes."

HackArena 2.0

Generative & Agentic AI

ModelRouter

github.com/dynamicip/modelrouter

FUTURE SCOPE

✓ MVP delivered ○ Vector search ○ Provider health ○ Managed SaaS